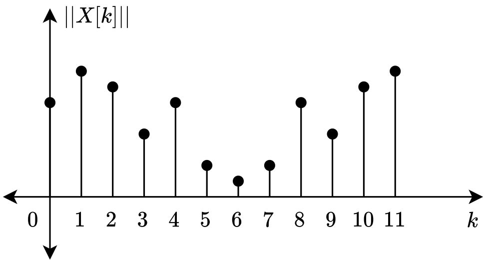
Converting FFT labels to frequency labels

Step - 0 Draw the signal

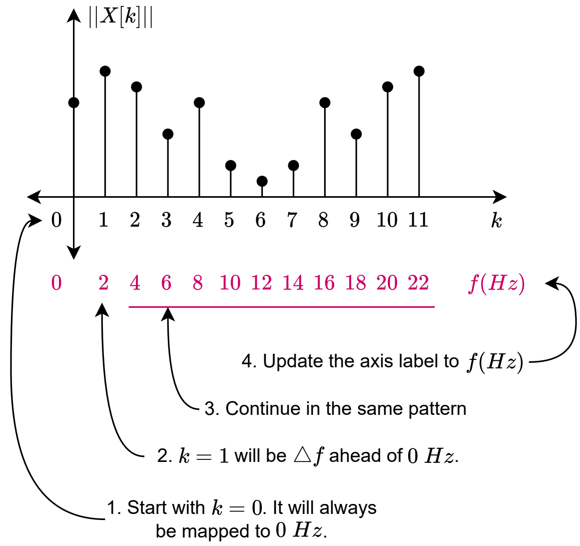


Step - 1 $\triangle f = \frac{f_s}{N}$

Let, $f_s = 24\ Hz$

Then, $\triangle f = \frac{24}{12} = 2Hz/sample$

Step - 2



1. Remember that unique information is only available till $\frac{f_s}{2}$, the Nyquist rate.

2. This ensures that there are only $\text{floor}(\frac{N}{2} + 1)$ usable samples in the spectrum. For a spectrogram we retain only the unique samples.

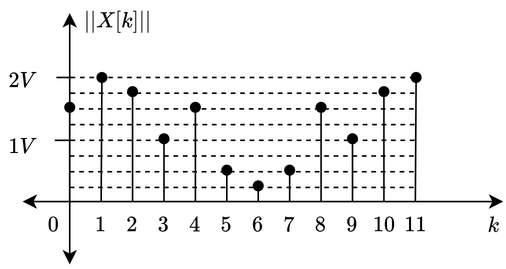
3. This means for a spectrogram, the frequency resolution is $\text{floor}(\frac{N}{2} + 1)$.

4. For real signals, all frequency values except the DC value ($0Hz$ component) are halved (half is distributed in the redundant portion).

5. So in case you are asked to manipulate the spectrum in preparation of a spectrogram, you have to double all the components except the DC value.

Preparing a regular FFT for standard spectrogram representation

Step - 0 Draw the signal. Note the amplitude levels.

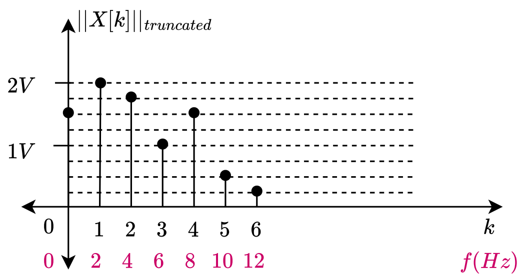


Step - 1 Find the number of unique samples using $\text{floor}(\frac{N}{2} + 1)$

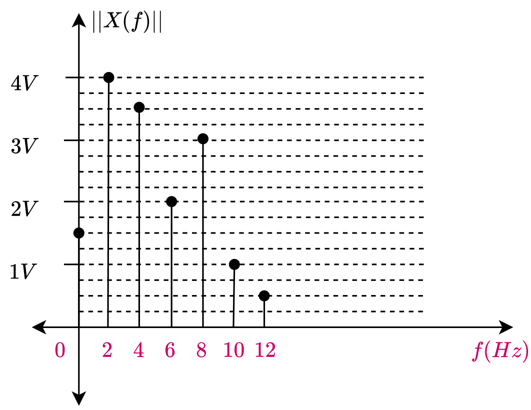
For this example, $N_{unique} = 7$.

Then, last index, $k' = 7 - 1 = 6$

Step - 2 Retain up to $k = k'$ and convert $x-$axis to Hz, $f_s = 24Hz$



Step - 3 Except for the $f = 0Hz$, double all the amplitudes.



Step - 4 You maybe asked to convert the amplitudes to dB.

In that case use this formula, $dB = 20log_{10}\frac{||X(f)||}{V_{ref}}$

You will be provided the value of V_{ref} in such scenarios.